

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 19)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Complete Table 1 to show the environmental information that can be collected by satellite sensors detecting each type of radiation.

[3 marks]

Table 1

Radiation detected	Environmental information collected	
Reflected visible light		
Infrared emitted by the Earth's surface		
Reflected radar microwaves emitted by the satellite		

0 1 . 2

Explain why radar surveys are not interrupted by cloudy weather.

[1 mark]

0 1 . 3

Explain **two** advantages of satellite surveys over ground surveys for monitoring deforestation.

[2 marks]

0 1 . 4

Describe how motion-sensitive cameras and image databases could be used together to estimate the size of a tiger population.

[4 marks]

10

02 . 1

Complete Table 1 to compare natural systems with typical linear human systems.

[4 marks]

Table 1

Feature	Natural systems	Linear human systems
Energy source		
Use of materials		
Nature of wastes		
Temperature of processes		

02 . 2

Explain why natural systems can be sustained using a relatively small number of elements.

[2 marks]

0 2 . 3

Explain why the human use of minerals makes them difficult to recover and re-use.

[2 marks]

0 2 . 4

Explain why toxic metals in electronic products cause problems that the wastes of natural systems do not.

[2 marks]

10

0 3 . 1

Explain why tropical rainforests have such high biodiversity.

[4 marks]

03 . 2

Explain why rainforest species may be particularly vulnerable to environmental change.

[2 marks]

03 . 3

Explain why rainforest plants and animals such as frogs are a rich source of medicines.

[2 marks]

03 . 4

Explain the importance of tropical rainforest for the hydrological cycle and for soils.

[2 marks]

10

04

Table 1 shows the estimated population of blue whales and the number killed by whaling in selected years.

Table 1

Year	1900	1930	1960	1986	2020
Estimated population	240 000	100 000	5 000	2 000	15 000
Whales killed that year	200	30 000	1 200	0	0

04 . 1

Calculate the percentage decline in the blue whale population between 1900 and 1986.

[1 mark]

Answer

%

04 . 2

Explain why whale populations were so easily over-exploited.

[3 marks]

04 . 3

Describe how the International Whaling Commission regulates whaling.

[4 marks]

04 . 4

Explain why the ITTO has been less successful than the IWC in achieving sustainable exploitation.

[2 marks]

10

0 5 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

2

0 5 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

0 5 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

5

0 6 . 1

Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

Square-mesh escape panels

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Minimum and maximum catchable sizes

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No-take zones

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Closed seasons

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06 . 2 Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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10

07 . 1

Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

07 . 2

Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

07 . 3

Explain why culling or removing species may be necessary in habitat management.

[2 marks]

10

08 . 1

Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

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08 . 2

For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Introduced competitor

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Introduced predator

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Introduced pathogen

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08 . 3

Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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10

09 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

09 . **1** Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1

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2

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3

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09 . **2** Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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09 . **3** Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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10

10 . 1

Evaluate the extent to which the principles of the circular economy can be applied to **three** of the following: water supplies, mineral supplies, waste management, pollution control, energy supplies, food production.

[9 marks]

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Extra space

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10 . **2** State what is meant by **industrial symbiosis**.

[1 mark]

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10

Write an essay on one of the following topics.

1	1	.	1
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Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

END OF QUESTIONS